

On hyperbolic approximations for a class of dispersive and diffusive-dispersive equations

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Joint work with
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The problem PDE

We consider for some interval $I \subset \mathbb{R}$ the scalar initial-value problem

$$\begin{aligned} u_t + f(u)_x &= \varepsilon u_{xx} + \gamma u_{xxx} \quad \text{in } I_T := I \times (0, T), \quad T > 0, \\ u(\cdot, 0) &= u_0(x). \end{aligned}$$

where

- $u(x, t)$ is the unknown.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth flux function.
 - $\gamma \in \mathbb{R}, \varepsilon \geq 0$.
- ⇒ Contains several models of wide interest: KdV, mKdV, KdV-Burgers, mKdV-Burgers, Gardner equation, etc.
- ⇒ Toy model to describe phase transition dynamics when viscous and capillary effects are taken into account.

The problem PDE - purely dispersive case

We consider for some interval $I \subset \mathbb{R}$ the scalar initial-value problem

$$\begin{aligned} u_t + f(u)_x &= \gamma u_{xxx} \quad \text{in } I_T := I \times (0, T), \quad T > 0, \\ u(\cdot, 0) &= u_0(x). \end{aligned}$$

$$\begin{aligned} \text{KdV eq:} \quad & f(u) = u^2/2, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + uu_x + \gamma u_{xxx} = 0, \\ \text{mKdV eq:} \quad & f(u) = u^3, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + 3u^2 u_x + \gamma u_{xxx} = 0, \\ \text{Gardner eq:} \quad & f(u) = u^3 + \mu u^2, \quad \gamma < 0 \quad \Rightarrow \quad u_t + (3u^2 + 2\mu u) u_x + \gamma u_{xxx} = 0. \end{aligned}$$

- Smooth solitary waves
- Dispersive shockwaves in the low dispersion limit $\gamma \rightarrow 0$.

The problem PDE - dissipative/dispersive case

We consider for some interval $I \subset \mathbb{R}$ the scalar initial-value problem

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KdV-B eq: $f(u) = u^2/2, \quad \varepsilon > 0, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + uu_x + \gamma u_{xxx} = \varepsilon u_{xx},$

mKdV-B eq: $f(u) = u^3, \quad \varepsilon > 0, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + 3u^2 u_x + \gamma u_{xxx} = \varepsilon u_{xx}.$

- Smooth solitary waves
- Possible undercompressive shocks in the sharp-interface limit.

Lower order formulations-approximations

- Second-order elliptic relaxation [Corli & Rohde 2012]

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x + \varepsilon u_{xx}^\alpha, \\ -\gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha), \quad 1 \ll \alpha. \end{aligned}$$

- Second-order parabolic relaxation [Corli, Rohde & Schleper 2014]

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x + \varepsilon u_{xx}^\alpha, \\ \beta c_t^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha), \quad 0 < \beta \ll 1. \end{aligned}$$

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- Can we apply a simple wave-type relaxation ?

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x + \varepsilon u_{xx}^\alpha, \\ \beta \textcolor{red}{c}_{tt}^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha). \end{aligned}$$

Plan of presentation

- 1 The 'hyperbolized' model
- 2 Model properties
- 3 Numerical results

First order hyperbolic relaxation (no dissipation)

1 Start from

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$$w^\alpha = c_t^\alpha, \quad p^\alpha = c_x^\alpha,$$

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$$\begin{aligned}u_t^\alpha + f(u^\alpha)_x + \alpha (u^\alpha - c^\alpha)_x &= 0, \\ \beta w_t^\alpha - \gamma p_x^\alpha &= \alpha (u^\alpha - c^\alpha), \\ c_t^\alpha &= w^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0.\end{aligned}$$

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$$w^\alpha = c_t^\alpha, \quad p^\alpha = c_x^\alpha, \quad \psi^\alpha = \alpha (u^\alpha - c^\alpha).$$

Then we obtain

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x + \alpha (u^\alpha - c^\alpha)_x &= 0, & u_t^\alpha + f(u^\alpha)_x + \psi_x^\alpha &= 0, \\ \beta w_t^\alpha - \gamma p_x^\alpha &= \alpha (u^\alpha - c^\alpha), & \Rightarrow \quad \beta w_t^\alpha - \gamma p_x^\alpha &= \psi^\alpha, \\ c_t^\alpha &= w^\alpha, & \psi_t^\alpha + \alpha f(u^\alpha)_x + \alpha \psi_x^\alpha &= -\alpha w^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0. & p_t^\alpha - w_x^\alpha &= 0. \end{aligned}$$

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$\Rightarrow$

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Hyperbolicity

We write the homogeneous system in quasilinear form

$$\mathbf{U}_t^\alpha + \mathbf{Df}(\mathbf{U}^\alpha) \mathbf{U}_x^\alpha = 0.$$

where the Jacobian is given by

$$\mathbf{Df}(\mathbf{U}) = \begin{pmatrix} f'(u) & 1 & 0 & 0 \\ \alpha f'(u) & \alpha & 0 & 0 \\ 0 & 0 & 0 & -\gamma/\beta \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

The eigenvalues of $\mathbf{Df}(\mathbf{U})$ compute as the real numbers $\lambda_1, \dots, \lambda_4$ given by

$$\lambda_1 = -\sqrt{\gamma/\beta}, \quad \lambda_2 = 0, \quad \lambda_3 = \sqrt{\gamma/\beta}, \quad \lambda_4 = f'(u) + \alpha.$$

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$\Rightarrow$ System is hyperbolic **only if** $\gamma/\beta > 0$

The mistake...

... Comes from the very beginning

$$\begin{aligned}u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x, \\ \beta c_{tt}^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha).\end{aligned}$$

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The fix

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\operatorname{sgn}(\gamma) \alpha (u^\alpha - c^\alpha)_x, \\ \beta c_{tt}^\alpha - |\gamma| c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha). \end{aligned}$$

Second attempt hyperbolic system

Now the system writes as

$$\begin{aligned}u_t^\alpha + (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= 0, \\ \psi_t^\alpha + \alpha (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= -\alpha w^\alpha, \\ w_t^\alpha - \beta^{-1} |\gamma| p_x^\alpha &= \beta^{-1} \psi^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0\end{aligned}$$

the Jacobian is now given by

$$\mathbf{Df}(\mathbf{U}) = \begin{pmatrix} f'(u) & \operatorname{sgn}(\gamma) & 0 & 0 \\ \alpha f'(u) & \alpha \operatorname{sgn}(\gamma) & 0 & 0 \\ 0 & 0 & 0 & -|\gamma|/\beta \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

The eigenvalues of $\mathbf{Df}(\mathbf{U})$ compute as the real numbers $\lambda_1, \dots, \lambda_4$ given by

$$\lambda_1 = -\sqrt{|\gamma|/\beta}, \quad \lambda_2 = 0, \quad \lambda_3 = \sqrt{|\gamma|/\beta}, \quad \lambda_4 = f'(u) + \alpha \operatorname{sgn}(\gamma).$$

Summary and honorable mentions

So far we have obtained a first-order hyperbolic system, regardless of the sign of γ .

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- ① can be derived from a variational principle.
- ② has an entropy/entropy flux pair such that

$$E^\alpha(\mathbf{U}) = \operatorname{sgn}(\gamma)F(u) + \frac{|\gamma|}{2}p^2 + \frac{1}{2\alpha}\psi^2 + \frac{\beta}{2}w^2, \quad F'(u) = f(u),$$

$$E_t^\alpha + \left(\frac{1}{2}f(u)^2 + f(u)\psi + \operatorname{sgn}(\gamma)\frac{\psi^2}{2} - |\gamma|pw \right)_x = 0$$

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What about the promised extension to the dissipative case ?

Viscous extension

Introduce the following relaxed system

$$u_t + f(u)_x + \operatorname{sgn}(\gamma)\alpha(u - c)_x = \varepsilon u_{xx},$$

$$\beta(c_{tt}) - |\gamma|c_{xx} = \alpha(u - c)$$

Viscous extension

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This did not work, full system was not dissipative!

Visco-elastic relaxation

Introduce the following relaxed system

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$$\beta(c_{tt} - \varepsilon c_{xxt}) - |\gamma|c_{xx} = \alpha(u - c)$$

Define the variables

$$w = c_t - \varepsilon c_{xx}, \quad p = c_x, \quad \psi = \alpha(u - c),$$

Obtain

$$u_t + (f(u) + \operatorname{sgn}(\gamma)\psi)_x = \varepsilon u_{xx},$$

$$\psi_t + \alpha(f(u) + \operatorname{sgn}(\gamma)\psi)_x = -\alpha w + \varepsilon \psi_{xx},$$

$$w_t - \beta^{-1}|\gamma|p_x = \beta^{-1}\psi,$$

$$p_t - w_x = \varepsilon p_{xx}$$

Visco-elastic relaxation

$$\begin{aligned} \frac{d}{dt} \int_I E^\alpha(\mathbf{U}) dx &= -\varepsilon \int_I \left(\operatorname{sgn}(\gamma) f'(u) u_x^2 + |\gamma| p_x^2 + \frac{\psi_x^2}{\alpha} \right) dx \\ &\leq 0. \end{aligned}$$

Entropy stable for $\gamma f(u)$ increasing in u .

Visco-elastic relaxation

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Entropy stable for $\gamma f(u)$ increasing in u .

Compatible with **one of the** original system entropies. Indeed

$$u_t + f(u)_x = \varepsilon u_{xx} + \gamma u_{xxx}$$

admits (among others) the entropy

$$E[u] = \operatorname{sgn}(\gamma) F(u) + \frac{|\gamma| u_x^2}{2}$$

satisfying

$$\frac{d}{dt} \int_I E[u] dx = -\varepsilon \int_I \left((\operatorname{sgn}(\gamma) f'(u) u_x^2 + |\gamma| u_{xx}^2) \right) dx$$

1 The 'hyperbolized' model

2 Model properties

3 Numerical results

Semi-Implicit Lax Wendroff - Intercell values

In order to solve the hyperbolic-parabolic system

$$\mathbf{U}_t^\alpha + \mathbf{f}(\mathbf{U}^\alpha)_x = \mathbf{S}\mathbf{U}^\alpha + \mathbf{D}\mathbf{U}_{xx}^\alpha$$

We use a semi-implicit Richtmeyer-Lax-Wendroff method.

1) We first evaluate the intercell values $\mathbf{U}_{i\pm\frac{1}{2}}^{n+\frac{1}{2}}$ via

$$\begin{aligned}\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} &= \frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i+1}^n) - \frac{\Delta t}{2\Delta x} (\mathbf{f}(\mathbf{U}_{i+1}^n) - \mathbf{f}(\mathbf{U}_i^n)) + \frac{\Delta t}{2} \mathbf{S}\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}}, \\ \mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} &= \frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i-1}^n) - \frac{\Delta t}{2\Delta x} (\mathbf{f}(\mathbf{U}_i^n) - \mathbf{f}(\mathbf{U}_{i-1}^n)) + \frac{\Delta t}{2} \mathbf{S}\mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}},\end{aligned}$$

whose values can be computed explicitly as

$$\mathbf{U}_{i\pm\frac{1}{2}}^{n+\frac{1}{2}} = \left(\mathbf{I} - \frac{\Delta t}{2} \mathbf{S} \right)^{-1} \left(\frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i\pm 1}^n) \mp \frac{\Delta t}{2\Delta x} (\mathbf{f}(\mathbf{U}_{i\pm 1}^n) - \mathbf{f}(\mathbf{U}_i^n)) \right).$$

Semi-Implicit Lax Wendroff - Update and parabolic terms

The solution of the hyperbolic first-order part of the system is then given by

$$\mathbf{U}_i^* = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x} \left(\mathbf{f} \left(\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \mathbf{f} \left(\mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} \right) \right) + \frac{\Delta t}{2} \mathbf{S} \left(\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} + \mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} \right).$$

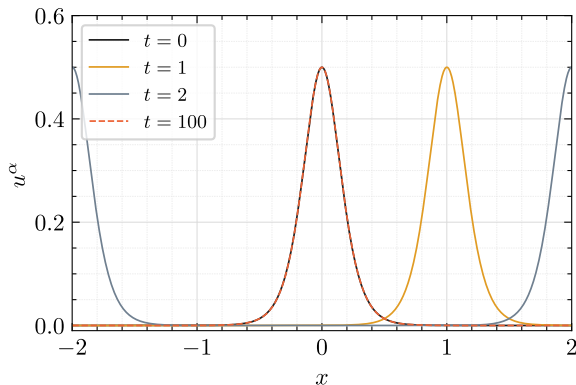
Finally, the solution at the new time, accounting for the second-order terms (present for $\varepsilon > 0$) is

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^* + \frac{\Delta t}{\Delta x^2} \mathbf{D} \left(\mathbf{U}_{i+1}^* - 2\mathbf{U}_i^* + \mathbf{U}_{i-1}^* \right).$$

KdV 1-Soliton solution

Target problem

$$\begin{cases} u_t + 6uu_x = \gamma u_{xxx}, \\ u(x, 0) = \frac{V}{2} \operatorname{sech}^2 \left(\sqrt{\frac{V}{4|\gamma|}} x \right), \\ \text{Periodic BC.} \end{cases}$$



$I_c = [-2, 2]$ with $N = 1000$ cells, $\gamma = -10^{-2}$, $\alpha = 10^3$ and $\beta = 10^{-6}$. Wave velocity $V = 1$ and final time is $T = 100$ (25 periods).

KdV 2-Soliton solution

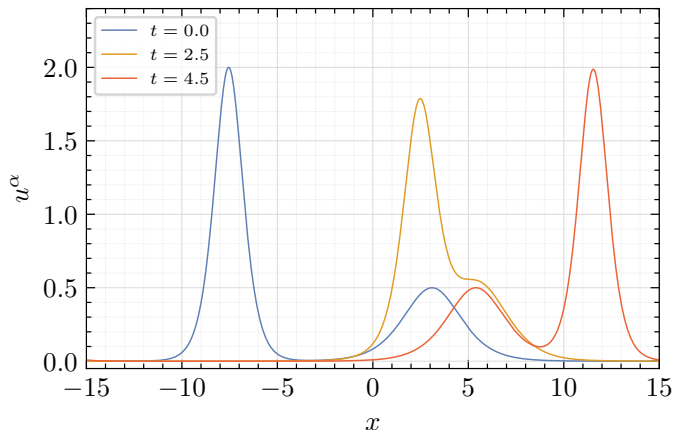
Target problem

$$\left\{ \begin{array}{l} u_t + 6uu_x = -u_{xxx}, \\ u(x, 0) = \frac{2(V_1 - V_2) (V_1 \cosh^2 \theta_2(x) + V_2 \sinh^2 \theta_1(x))}{((\sqrt{V_1} + \sqrt{V_2}) \cosh(\theta_1(x) - \theta_2(x)) + (\sqrt{V_1} - \sqrt{V_2}) \cosh(\theta_1(x) + \theta_2(x)))^2}, \\ \text{where } \theta_j(x) = \sqrt{V_j}(x - x_j)/2, \quad j = 1, 2 \\ \text{Periodic BC.} \end{array} \right.$$

KdV 2-Soliton solution

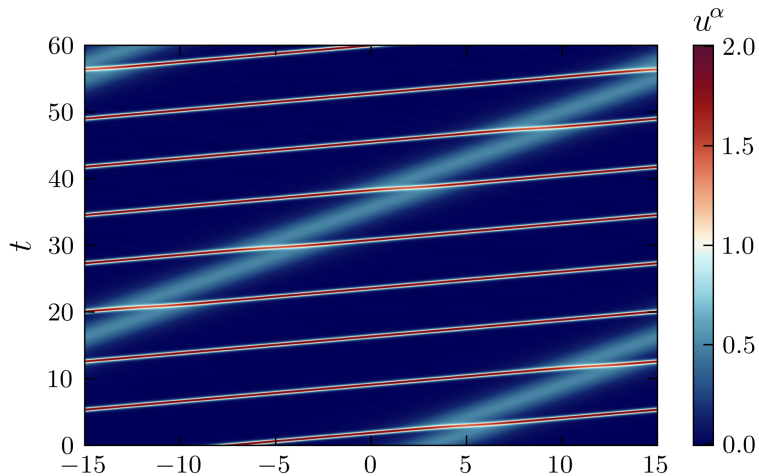
Target problem

$$\begin{cases} u_t + 6uu_x = -u_{xxx}, \\ u(x, 0) = \dots \\ \text{Periodic BC.} \end{cases}$$



$I_c = [-15, 15]$ with $N = 2000$ cells, $\alpha = 4000$ and $\beta = 10^{-6}$. Final time is $T = 60$.

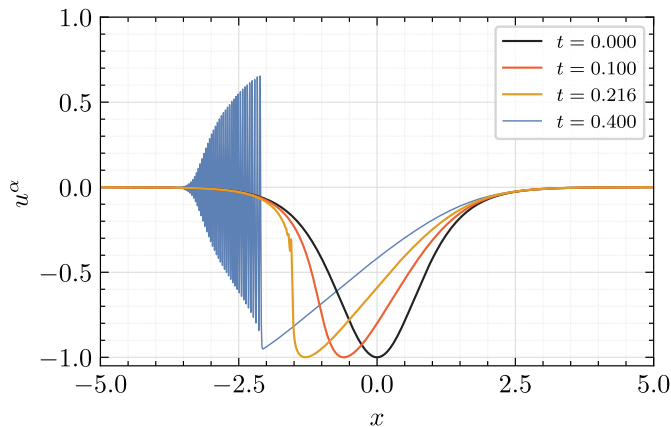
KdV 2-Soliton solution $(x - t)$ diagram



KdV negative hump DSW

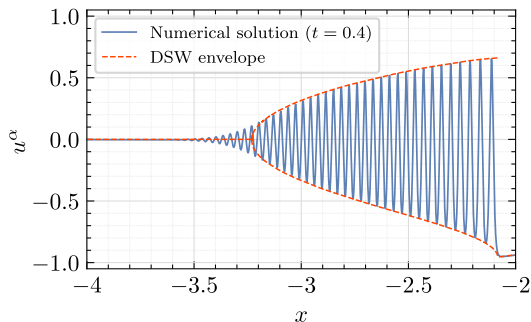
Target problem

$$\begin{cases} u_t + 6uu_x = \gamma u_{xxx}, \\ u(x, 0) = -\text{sech}^2(x), \\ \text{Periodic BC.} \end{cases}$$

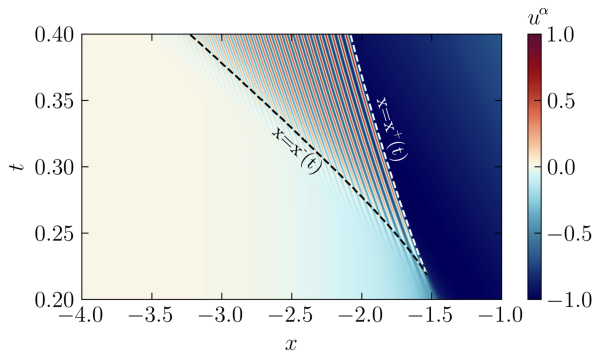


$I_c = [-5, 5]$ with $N = 8000$ cells, $\gamma = -10^{-4}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

KdV negative hump DSW



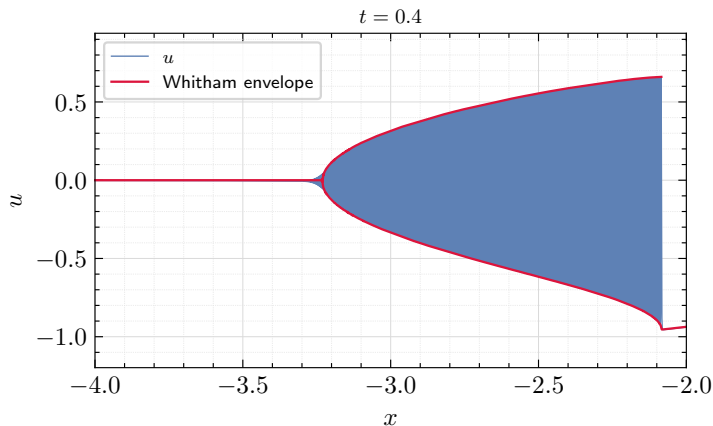
(a) Comparison of the numerical solution with the Whitham envelope for the target problem.



(b) $x - t$ diagram of the numerical solution at the onset of the DSW formation.

$I_c = [-5, 5]$ with $N = 8000$ cells, $\gamma = -10^{-4}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

KdV negative hump DSW - low dispersion limit

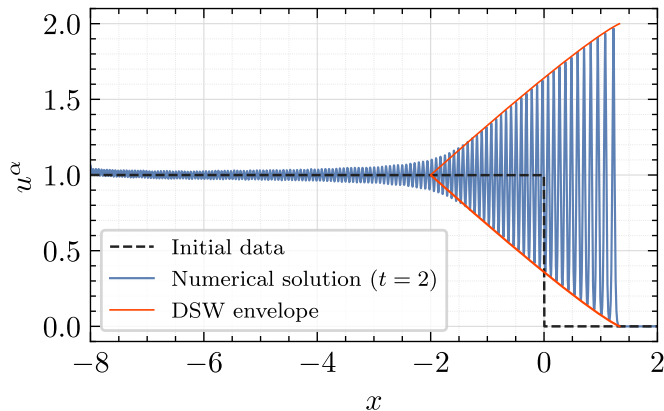


$I_c = [-5, 5]$ with $N = 60000$ cells, $\gamma = -10^{-6}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

KdV Riemann problem

Target problem

$$\begin{cases} u_t + uu_x = \gamma u_{xxx}, \\ u(x, 0) = \begin{cases} 1 & \text{if } x < 0, \\ 0 & \text{if } x \geq 0. \end{cases} \end{cases}$$



$I_c = [-8, 2]$ with $N = 10000$ cells, $\gamma = -10^{-4}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

KdV-Burgers Traveling wave front

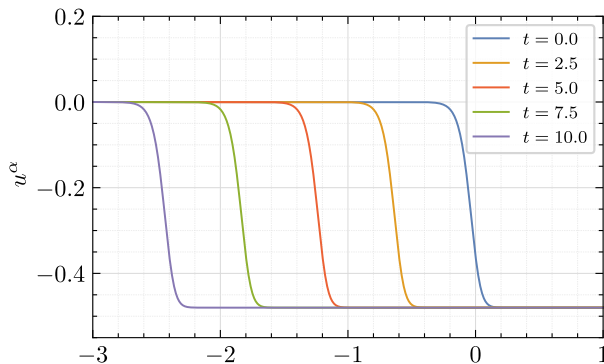
Target problem

$$\begin{cases} u_t + uu_x = \varepsilon u_{xx} + \gamma u_{xxx}, \\ u(x, 0) = \frac{-3\varepsilon^2}{25\gamma} \left(2 + 2 \tanh \left(\frac{\varepsilon x}{10\gamma} \right) + \operatorname{sech}^2 \left(\frac{\varepsilon x}{10\gamma} \right) \right). \end{cases}$$

KdV-Burgers Traveling wave front

Target problem

$$\begin{cases} u_t + uu_x = \varepsilon u_{xx} + \gamma u_{xxx}, \\ u(x, 0) = \dots \end{cases}$$



$I_c = [-3, 1]$ with $N = 2000$ cells, $\gamma = 10^{-4}$, $\alpha = 10^4$ and $\beta = 10^{-6}$. Solution is shown at different times up to $T = 10$.

Convergence in the relaxation parameters

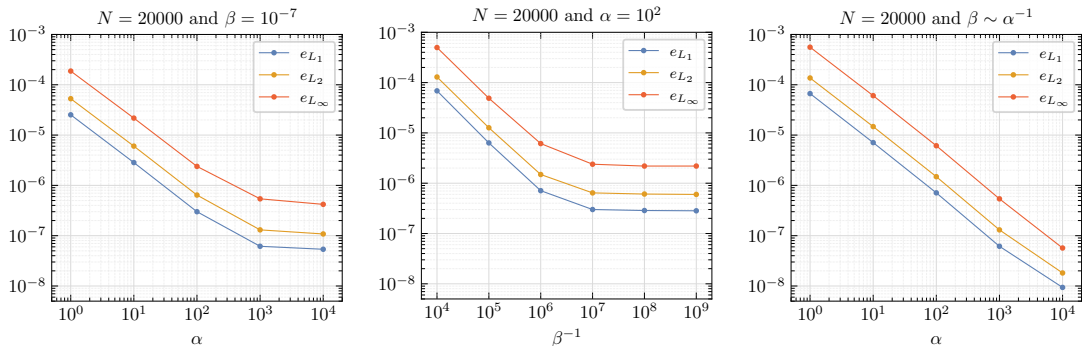


Figure 6: Convergence plots in different error norms. Left: Error decay and saturation for fixed β and $\alpha \rightarrow \infty$. Middle: Error decay and saturation for fixed β and $\alpha \rightarrow \infty$. Right: Linear error decay for $\beta = \gamma\alpha^{-1}$ and $\alpha \rightarrow \infty$.

mKdV-Burgers Undercompressive shock

$$\begin{cases} u_t + 3u^2u_x = \varepsilon u_{xx} - \gamma u_{xxx}, \\ u(x, 0) = \frac{1}{2} (u_+ + u_- - |u_+ - u_-| \tanh(Ax/\sqrt{\gamma})) , \end{cases}$$

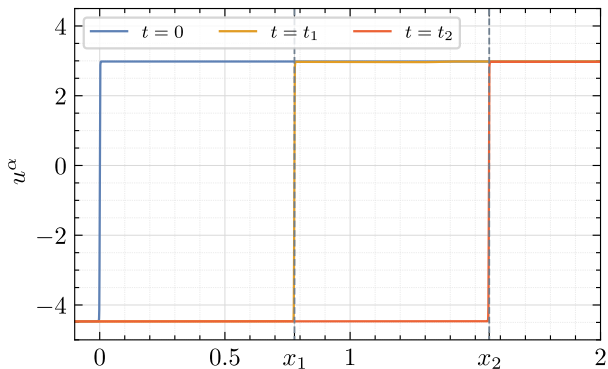
$$u_- = -\sqrt{\frac{2\varepsilon^2}{\gamma}}, \quad u_+ = -u_- - \sqrt{\frac{2\varepsilon^2}{9\gamma}}, \quad A = \frac{u_- - u_+}{2\sqrt{2}}.$$

For these values, in the zero dispersion/dissipation limit, an undercompressive shock is obtained since the corresponding shock is subsonic both ahead of and behind the wave:

$$\begin{aligned} u_- &\simeq -4.47, & u_+ &\simeq 2.97, & s &= \frac{u_+^3 - u_-^3}{u_+ - u_-} \\ \lambda_- &\simeq 59.94, & \lambda_+ &\simeq 26.46, & s &\simeq 15.52 \end{aligned}$$

parameters: $\varepsilon = 10^{-2}$, $\gamma = 10^{-5}$

mKdV-Burgers Undercompressive shock



$I_c = [-0.1, 2]$ with $N = 10000$ cells, $\gamma = 10^{-5}$, $\alpha = 10^3$ and $\beta = 10^{-11}$. Solution is shown at different times up to $T = 10$.

Conclusions - Perspectives

Summary

- ① A new first-order hyperbolic model for general conservation laws with dissipative/dispersive regularization is presented.
- ② Convergence of its solutions towards those of the original model is proven for sufficiently regular solutions.
- ③ The numerical results shows applicability of the approach for several PDEs.

Perspectives

- ① Performance - comparison with existing methods ? Is it worth it ?
- ② Fully hyperbolic model ?
- ③ Vanishing-viscosity solution convergence for hyperbolized system ?

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Side (but important) note

- Global well-posedness and convergence of solutions of the relaxed system towards those of the original model is proven for sufficiently regular solutions.



Thank you for your attention !

[1] Barthwal, Rahul, Firas Dhaouadi, and Christian Rohde. "On hyperbolic approximations for a class of dispersive and diffusive-dispersive equations." arXiv preprint arXiv:2512.04882 (2025).

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Convergence of solutions (also works for $\varepsilon = 0$)

Theorem (Main result)

Let $u_0 \in H^6(I)$, and suppose $|f(u)'|, |f(u)''| < L$. We assume that $u \in L^2(0, T; H^3(I))$ with $u_{tt} \in L^2(I_T)$ and $\|u\|_{L^\infty(0, T; H^4(I))} < \infty$ is a classical solution of

$$u_t + f(u)_x = \varepsilon u_{xx} + \gamma u_{xxx}, \quad u(x, 0) = u_0(x)$$

Consider for $\beta = \alpha^{-1}$, a sequence $\{\mathbf{U}^\alpha = (u^\alpha, \psi^\alpha, w^\alpha, p^\alpha)^\top\}_{\alpha>0}$ of weak entropy solutions of

$$\mathbf{U}_t^\alpha + \mathbf{f}(\mathbf{U}^\alpha)_x = \mathbf{S}(\mathbf{U}^\alpha) + \mathbf{D}\mathbf{U}^\alpha_{xx} \quad + \quad \text{Well-prepared IC}$$

Then there is a constant $C = C(L, \delta, u) \geq 0$ that is independent of α such that

$$\|u^\alpha - u\|_{L^\infty(0, T; L^2(I))} + \|p^\alpha - u_x\|_{L^\infty(0, T; L^2(I))} \leq C\alpha^{-1/2}.$$

Convergence of solutions - proof elements

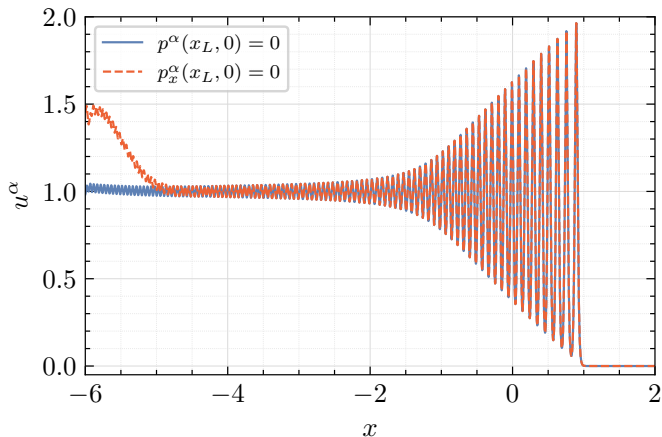
- 1 Write the original system in a similar form as its approximation.

$$\begin{aligned}
 u_t + f(u)_x + \operatorname{sgn}(\gamma)\psi_x &= 0, \\
 \frac{1}{\alpha}\psi_t + f(u)_x + \operatorname{sgn}(\gamma)\psi_x &= -w - \frac{1}{\alpha}|\gamma|u_{xxt}, \\
 \beta w_t - |\gamma|p_x &= \psi + \beta u_{tt}, \\
 p_t - w_x &= 0.
 \end{aligned} \tag{1}$$

- 2 Use Relative entropy framework.
- 3 Grönwall type arguments.

"Homogeneous Neumann" Boundary conditions

$$\begin{aligned} u_{\mathbf{x}}(x_L, t) &= u_{\mathbf{x}}(x_R, t) = 0, \\ \psi_{\mathbf{x}}(x_L, t) &= \psi_{\mathbf{x}}(x_R, t) = 0, \\ w_{\mathbf{x}}(x_L, t) &= w_{\mathbf{x}}(x_R, t) = 0, \\ p(x_L, t) &= p(x_R, t) = 0. \end{aligned}$$



Comparison of suggested "Neumann" boundary conditions with classic Neumann boundary

Dissipative extension

- 1 Simply insert the parabolic term back to the system

$$\begin{aligned} u_t^\alpha + (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= \varepsilon u_{xx}^\alpha, \\ \psi_t^\alpha + \alpha (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= -\alpha w^\alpha, \\ w_t^\alpha - \beta^{-1} |\gamma| p_x^\alpha &= \beta^{-1} \psi^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0 \end{aligned}$$

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- 2 Derive the energy equation

$$\begin{aligned} E_t^\alpha + \left(\frac{1}{2} f(u^\alpha)^2 + f(u^\alpha)\psi + \operatorname{sgn}(\gamma) \frac{\psi^{\alpha 2}}{2} - |\gamma| p^\alpha w^\alpha \right)_x &= \varepsilon f(u^\alpha) u_{xx}^\alpha. \\ &= (\varepsilon f(u^\alpha) u_x^\alpha)_x - f'(u^\alpha) (u_x^\alpha)^2 \end{aligned}$$

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Dissipated entropy for increasing flux functions $f(u)$.